

TOTUS TUUS KEELY

MECHATRONICS ENGINEERING | MECHANICAL DESIGN | AUTOMATION | ROBOTICS

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PROFESSIONAL SUMMARY

Mechatronics Engineering student graduating May 2027 with three engineering internships in industrial equipment design, automated machinery and manufacturing operations. Hands-on experience with SolidWorks, robotics, embedded systems, validation testing and design for manufacturability, plus leadership of a 50-person college news organization.

EDUCATION

Northern Illinois University | B.S. in Mechatronics Engineering | Expected May 2027

GPA: 3.82 | Dean's List | DeKalb, IL | **Coursework:** Robotics and Automation | AI for Robotics | Design of Mechatronics Systems

Credentials: FAA Part 107 Remote Pilot Certificate; NVIDIA Fundamentals of Deep Learning; LinkedIn Learning - SOLIDWORKS 2022 Essential Training

TECHNICAL SKILLS

CAD & Documentation: SolidWorks, large assemblies, sheet metal, weldments, drawings, PDM, Autodesk Inventor, BOMs

Automation & Programming: RoboDK, robot kinematics, motion planning, collision checking, Python, C++, MATLAB

Embedded & Manufacturing: Raspberry Pi 5, ESP32, Arduino Nano RP2040, DFM, validation testing, CNC routers, mills, lathes

ENGINEERING EXPERIENCE

Precision Quincy Inc. | Mechanical Engineering Intern | South Beloit, IL | May 2026 - Aug. 2026

- Designed an ~100-part industrial spray booth in SolidWorks, including sheet metal, weldments, drawings, DXFs and BOMs.
- Evaluated airflow, anchoring and hinge strength and incorporated ~10 design changes.
- Completed a 500-sample capacitor-discharge weld study across 5 variables.
- Conducted destructive laser-vs.-MIG shear testing and trained manufacturing personnel on laser welding.

Pregis IQ | Mechanical Engineering Intern | Aurora, IL | May 2025 - Aug. 2025

- Designed approximately 50 parts and 6 SolidWorks assemblies for automated packaging equipment, created manufacturing drawings and managed design data in SolidWorks PDM.
- Developed a custom test fixture through 4 prototype iterations and test procedures demonstrating approximately 40% greater machine efficiency; authored 3 user and technical manuals.

American Ordnance | Operations Engineering Intern | Middletown, IA | June 2024 - Aug. 2024

- Reduced rework on one production line from 10% to 5% and built Excel databases and hours-per-unit tools that documented a 30% summer improvement.

LEADERSHIP & SELECTED PROJECTS

Northern Star Media | Editor in Chief | DeKalb, IL | May 2026 - Present

- Lead approximately 50 staff members with 14 direct reports and oversee at least 8 digital stories daily, 3 print editions per semester and a daily five-story newsletter.
- Built a Microsoft Power Automate workflow integrating Microsoft Lists to send personalized interview-scheduling emails, reducing manual administrative work.

Industrial Robot Pick-and-Place Simulation | Robotics and Automation | Spring 2026

- Programmed a 6-axis HIWIN robot in RoboDK using the Python API for frames, 4x4 transformations, kinematics, motion planning, rotation-matrix validation and collision checking.

Autonomous Quadruped Navigation Robot | Fundamentals of Mechatronics | Fall 2025

- Co-developed Raspberry Pi 5 and ESP32-based navigation using servo-mounted ultrasonic sensing for object avoidance, wall detection and wall following.